

MATH119: Big Data
Homework 2
Steven Finnini

1) a) let $\sigma(x) = \frac{1}{1+e^{-x}}$ show that $\sigma'(x) = \sigma(x)[1-\sigma(x)]$

$$\frac{\partial \sigma}{\partial x} \left[\sigma(x) = \frac{1}{1+e^{-x}} \right]$$

$$\frac{\partial \sigma}{\partial x} = (1+e^{-x})^{-1} \cdot \frac{\partial}{\partial x} (1+e^{-x})^{-1}$$

$$\frac{\partial \sigma}{\partial x} = -(1+e^{-x})^{-2} \cdot -e^{-x}$$

$$= (1+e^{-x})^{-2} \cdot e^{-x}$$

$$\sigma'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$\sigma'(x) = \sigma(x)[1-\sigma(x)]$$

$$1-\sigma(x) = 1 - \frac{1}{1+e^x}$$

$$1-\sigma(x) = \frac{1+e^x}{1+e^x} - \frac{1}{1+e^x}$$

$$1-\sigma(x) = \frac{e^{-x}}{1+e^x}$$

$$\sigma'(x) = 1 - \sigma(x) \therefore$$

b) previous result + chain rule $\rightarrow$ derive gradient of the log likelihood for logistic regression using negative log-likelihood

- observations: (x_i, y_i) , $y_i \in \{0, 1\}$

- logistic regression models the probability that $y_i = 1$ as: $\mu_i = \sigma(w^T x_i)$

$$\mu_i = \frac{1}{1+e^{-w^T x_i}} \quad P(y_i | x_i, w) = \mu_i^{y_i} (1-\mu_i)^{1-y_i} \quad y_i = 1 \rightarrow \mu_i$$

- likelihood: $\mathcal{L}(w) = \prod_{i=1}^n \mu_i^{y_i} (1-\mu_i)^{1-y_i}$ $y_i = 0 \rightarrow 1-\mu_i$

$$\log[\mathcal{L}(w)] = \sum_{i=1}^n [y_i \log(\mu_i) + (1-y_i) \log(1-\mu_i)] \quad \nabla_w \mathcal{L}(w)$$

$$\mu_i = \sigma(w^T x_i) \quad \nabla_w \mathcal{L}_i = -[y_i \nabla_w \log(\mu_i) + (1-y_i) \nabla_w \log(1-\mu_i)]$$

Differentiate $\log(\mu_i)$

$$\nabla_w \log(\mu_i) = \frac{1}{\mu_i} \nabla_w \mu_i$$

$$\nabla_w \mu_i \quad \mu_i = \sigma(w^T x_i) \quad \nabla_w \mu_i = \sigma'(w^T x_i) \nabla_w (w^T x_i)$$

$$\sigma'(x) = \sigma(x)[1-\sigma(x)] \quad \sigma(w^T x_i) = \mu_i (1-\mu_i)$$

$$\nabla_w (w^T x_i) = x_i \quad \boxed{\nabla_w \mu_i = \mu_i (1-\mu_i) x_i}$$

$$\nabla_w \log(\mu_i) = \frac{1}{\mu_i} \mu_i (1-\mu_i) x_i$$

$$\boxed{\nabla_w \log(\mu_i) = (1-\mu_i) x_i}$$

Differentiate $\log(1-\mu_i)$

$$\nabla_w \log(1-\mu_i) = \frac{1}{1-\mu_i} \nabla_w (1-\mu_i)$$

$$\nabla_w (1-\mu_i) = -\nabla_w \mu_i$$

$$\nabla_w \log(1-\mu_i) = -\frac{1}{1-\mu_i} \nabla_w \mu_i$$

$$\nabla_w \log(1-\mu_i) = -\frac{\mu_i (1-\mu_i) x_i}{1-\mu_i}$$

$$\boxed{\nabla_w \log(1-\mu_i) = -\mu_i x_i}$$

$$\nabla_w \mathcal{L}_i = -[y_i \nabla_w \log(\mu_i) + (1-y_i) \nabla_w \log(1-\mu_i)]$$

$$\nabla_w \mathcal{L}_i = -[y_i (1-\mu_i) x_i + (1-y_i) (-\mu_i x_i)]$$

$$= -[y_i (1-\mu_i) + (1-y_i) (-\mu_i)] x_i$$

$$= -[y_i - y_i \mu_i + \mu_i - y_i \mu_i] x_i$$

$$\boxed{\nabla_w \mathcal{L}_i = (\mu_i - y_i) x_i}$$

$$\nabla_w \mathcal{L}(w) = \sum_{i=1}^n (\mu_i - y_i) x_i \quad \text{where } \mu_i = \sigma(w^T x_i)$$

$$\boxed{\nabla_w \mathcal{L}(w) = X^T (\mu - y)}$$

c) derive Hessian

$$H = X^T \delta X, \quad \text{where } \delta = \text{diag}(\mu_1(1-\mu_1), \dots, \mu_n(1-\mu_n))$$

Show $H \succeq 0$, H is positive semi-definite

$$\nabla_w \mathcal{L}(w) = \sum_{i=1}^n (\mu_i - y_i) x_i, \quad \mu_i = \sigma(w^T x_i)$$

$$\nabla_w \mu_i = \mu_i (1-\mu_i) x_i$$

$$H = \nabla_w^T \mathcal{L}(w) = \sum_{i=1}^n \mu_i (1-\mu_i) x_i x_i^T$$

$$\text{let } \delta = \text{diag}(\mu_1(1-\mu_1), \dots, \mu_n(1-\mu_n))$$

$$\boxed{H = X^T \delta X}$$

to show $H \succeq 0$, let z be any vector:

$$z^T H z = z^T X^T \delta X z$$

$$= (X z)^T \delta (X z)$$

$$= \sum_{i=1}^n \mu_i (1-\mu_i) (x_i^T z)^2$$

$$\text{since } 0 < \mu_i < 1, \mu_i (1-\mu_i) \geq 0, (x_i^T z)^2 \geq 0.$$

$$\text{Hence, } z^T H z \geq 0, \text{ for all } z.$$

$$\text{Therefore, } H \succeq 0.$$

2) derive normalization constant (Z) for 1-dimensional zero-mean Gaussian

$$P(x; \sigma^2) = \frac{1}{Z} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad \text{such that } P(x; \sigma^2) \text{ becomes a valid density.}$$

- probability density must integrate to 1 over all possible values.

$$\int_{-\infty}^{\infty} P(x; \sigma^2) dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{Z} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx = \frac{1}{Z} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx = 1$$

$$\boxed{Z = \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx} \quad \times Z \quad \times Z$$

$$I = \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx, \quad \text{then } I^2 = Z$$

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx \right)^2$$

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2\sigma^2} dy \right)$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} e^{-y^2/2\sigma^2} dx dy$$

$$e^a e^b = e^{a+b}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2\sigma^2} dx dy$$

because $x^2+y^2 \rightarrow$ convert to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta \quad \text{and} \quad x^2+y^2 = r^2$$

$$dx dy \rightarrow r dr d\theta$$

original x, y integral covers entire plane, so polar-coordinate bounds:

$$0 \leq r < \infty \quad \text{and} \quad 0 \leq \theta \leq 2\pi$$

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2\sigma^2} r dr d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2/2\sigma^2} dr$$

$$I^2 = 2\pi \int_0^{\infty} r e^{-r^2/2\sigma^2} dr \quad \int_0^{\infty} r e^{-r^2/2\sigma^2} dr$$

$$u = \frac{r^2}{2\sigma^2} = r^2 (2\sigma^2)^{-1}$$

$$du = \frac{2r}{2\sigma^2} dr = \frac{r}{\sigma^2} dr$$

$$\sigma^2 du = r dr$$

$$I^2 = 2\pi \int_0^{\infty} e^{-u} \sigma^2 du$$

$$I^2 = 2\pi \sigma^2 \int_0^{\infty} e^{-u} du$$

$$\int_0^{\infty} e^{-u} du = [-e^{-u}]_0^{\infty}$$

$$e^{-\infty} = 0, \quad e^0 = 1$$

$$= 0 - (-1) = 1$$

$$I^2 = 2\pi \sigma^2$$

$$\sqrt{I^2} = \sqrt{2\pi \sigma^2}$$

$$I = \sqrt{2\pi \sigma^2} = \sigma \sqrt{2\pi}$$